

JEE MAIN REVISION MASTER: BASICS OF ORGANIC CHEMISTRY

Subject: Chemistry | **Target Scope:** JEE Main Previous Years | **Empirical Base:** 295 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Basics Of Organic Chemistry — Core Principles & Governing Laws	178 PYQs	Core Principle
Very High	Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	106 PYQs	Applied Topics
Very High	Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes	21 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basics Of Organic Chemistry — Core Principles & Governing Laws	$\text{CH}_3\text{-CH=CH-CHO}$, $\quad _7$, $\quad =\frac{80 \times 90}{100}=72\text{gm}$
Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	-NHCOCH_3 , $\quad$ $R_f=\frac{\text{Distancetravelledbycompound}}{\text{Distancetravelledbysolvent}}$, $\quad$ NO_2
Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes	$R_f=\frac{\text{Distancetravelledbycompound}}{\text{Distancetravelledbysolvent}}$, $\quad$ $\wedge\text{-OH}$, $\quad \text{-NH}_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basics Of Organic Chemistry — Core Principles & Governing Laws (Very High Yield — 178 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Basics Of Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Basics Of Organic Chemistry PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Vapours of the liquid with higher boiling point condense before vapours of the liquid with lo..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "For the given molecule, "x", the preferred site for the attack of the electrophile is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : There are several conformers for n-butane. Out of those conformers, Statement II : As the di..." [Ans: N/A]

3.2 Basics Of Organic Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 106 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Reaction List-I Mechanism A. Williamson Synthesis I. Electrophilic addition B. Friedel Craft R..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The strongest conjugate acid will result from :..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Consider the following molecules/species : The correct order of carbon-oxygen double bond length is :..." [Ans: N/A]

3.3 Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes (Very High Yield — 21 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "Increasing order of electron withdrawing power of following functional groups is : a. -CN b. -COOH c. -NO₂ d. -I..." [Ans: N/A]
- [2026 Evening Shift] Q#4 (MCQ): "Given below are four compounds : (a) n-propyl chloride (b) iso-propyl chloride (c) sec-butyl chloride (d) neo-pentyl chloride Per..." [Ans: N/A]
- [2025 Morning Shift] Q#4 (MCQ): "How many different stereoisomers are possible for the given molecule?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 23rd January Morning Shift (Q#1)

Given below are two statements : Statement I : Sublimation is used for the separation and purification of compounds with low melting point. Statement II : The boiling point of a liquid increases as the external pressure is reduced. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are false **[Correct]**
- (B) Statement I is true but Statement II is false
- (C) Both Statement I and Statement II are true
- (D) Statement I is false but Statement II is true

Correct Answer: A

Solution Analysis: Statement I: False. Sublimation is used to separate/purify sublimable (volatile) solids —solids that pass directly from solid to vapour on heating. This depends on having sufficient vapour pressure in the solid state, not simply on having a low melting point . Statement II: False. The boiling point decreases when external pressure is reduced (principle of vacuum distillation). ■ Correct option: A (Both Statement I and Statement II are false).

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

Identify the correct statements : The presence of $-\text{NO}_2$ group in benzene ring A. activates the ring towards electrophilic substitutions. B. deactivates the ring towards electrophilic substitutions. C. activates the ring towards nucleophilic substitutions. D. deactivates the ring towards nucleophilic substitutions. Choose the correct answer from the options given below :

- (A) and D Only
- (B) and D Only **[Correct]**
- (C) and A Only
- (D) and C Only

Correct Answer: B

Solution Analysis: Presence of NO_2 group on the benzene ring deactivates the ring towards electrophilic substitution reactions because it shows -M effect (it withdraws electrons from the ring by resonance). Because the NO_2 group pulls electron density away, the ring becomes less reactive towards electrophiles. At the same time, the NO_2 group activates the ring towards nucleophilic substitution because the withdrawal of electrons makes the ring more suitable for attack by nucleophiles (electron-rich species). Ans. $\rightarrow$ B & C

Benchmark Problem 3: JEE Main 2025 (Online) 22nd January Morning Shift (Q#4)

In Carius method for estimation of halogens, 180 mg of an organic compound produced 143.5 mg of AgCl . The percentage composition of chlorine in the compound is _____. (Given : molar mass in g mol^{-1} of Ag :108, Cl :35.5)

Correct Answer: 20

Solution Analysis: To find the percentage composition of chlorine in an organic compound using the Carius method, we follow these steps: Calculate Millimoles of AgCl : Since 143.5 mg of AgCl is produced, and the molar mass of AgCl is 143.5 g/mol, the millimoles of AgCl is: Millimoles of $\text{AgCl} = (143.5 \text{ mg} / 143.5 \text{ mg/mmol}) = 1 \text{ mmol}$ Determine Millimoles of Cl : In AgCl , there is a 1:1 ratio of Ag to Cl , so the millimoles of Cl are equal to that of AgCl : Millimoles of $\text{Cl} = 1 \text{ mmol}$ Calculate Mass of Cl : Using the molar mass of Cl (35.5 g/mol), the mass of Cl is: $\text{Mass of Cl} = 1 \text{ mmol} \times 35.5 \text{ g/mol} = 35.5 \text{ mg}$